

SmartBranch 360

Secure Branch Office Network — Complete Project Report

Cisco AICTE Virtual Internship Program 2026 — Project 1: Networking + Packet Tracer + Python

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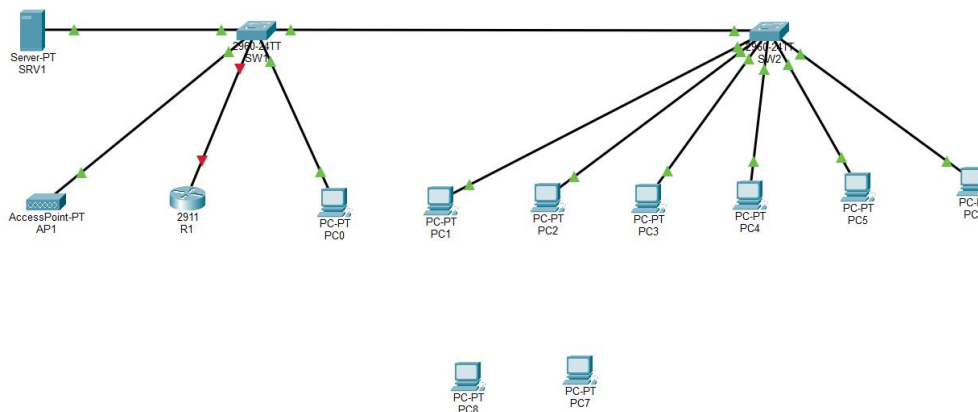
1. Objective

Design and build a small branch-office network in Cisco Packet Tracer supporting wired employees, guest Wi-Fi, an internal server, internet access via NAT, and secure out-of-band management — then validate the design with a Python assurance tool and prove resilience through five fault-injection/troubleshooting cycles, all with verified evidence.

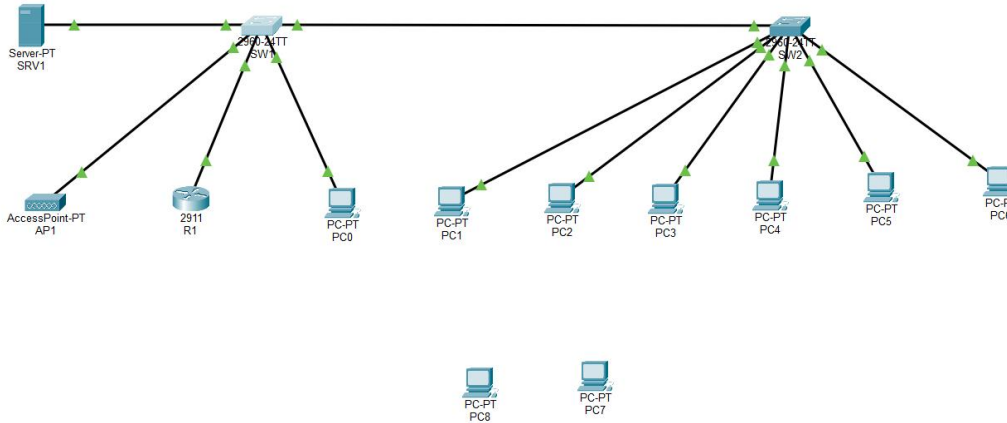
2. Topology Overview

The branch uses one ISR router (R1) performing router-on-a-stick inter-VLAN routing and NAT, a two-tier switch layer (SW1 distribution, SW2 access), one wireless access point (AP1) serving the guest SSID, one internal server (SRV1), and 9 endpoints (6 wired employee PCs, 1 wired management PC, 2 wireless guest devices).

- R1 — GigabitEthernet0/0 trunked to SW1 (sub-interfaces per VLAN); GigabitEthernet0/1 faces the Internet cloud, NAT-outside.
- SW1 — Distribution switch; trunks to R1, SW2, AP1; SRV1 and Management PC on access ports.
- SW2 — Access switch; 6 wired employee PCs on access ports; trunk uplink to SW1.
- AP1 — Broadcasts guest SSID 'SmartBranch-Guest' (WPA2-PSK/AES); wired uplink is an access port in VLAN 20.



Built topology in Packet Tracer: R1, SW1, SW2, AP1, SRV1, and all 9 endpoints connected.



All links verified up (green) after configuration — every port active and forwarding correctly.

3. VLAN and IP Address Plan

VLAN	Name	Subnet	Gateway	Address Method
10	EMPLOYEE	10.10.10.0/24	10.10.10.1	DHCP (.100–.200)
20	GUEST	10.10.20.0/24	10.10.20.1	DHCP (.50–.150), wireless
30	SERVER	10.10.30.0/24	10.10.30.1	Static (SRV1 = .10)
99	MANAGEMENT	10.10.99.0/24	10.10.99.1	Static (MgmtPC = .2)

4. Services Configured

- DHCP: per-VLAN pools on R1 for EMPLOYEE and GUEST; SERVER and MANAGEMENT use static addressing.
- DNS: forwarded to 8.8.8.8 by default; an internal DNS record was also added on SRV1 during Fault 4 troubleshooting (see Section 6).
- NAT: PAT (overload) on R1's outside interface (Gi0/1) for EMPLOYEE and GUEST subnets.
- Inter-VLAN routing: router-on-a-stick using dot1Q sub-interfaces Gi0/0.10 / .20 / .30 / .99 on R1.
- Wireless: AP1 broadcasts SSID 'SmartBranch-Guest' with WPA2-PSK/AES authentication, mapped to VLAN 20.

The screenshot shows a configuration window with the following fields and values:

- WEK Key: (empty)
- PSK Pass Phrase: Guest@123
- User ID: (empty)
- Password: (empty)
- Authentication Method: A dropdown menu is open, showing three options: AES (selected), AES, and TKIP.

AP1 wireless configuration — SSID and WPA2-PSK/AES authentication set.

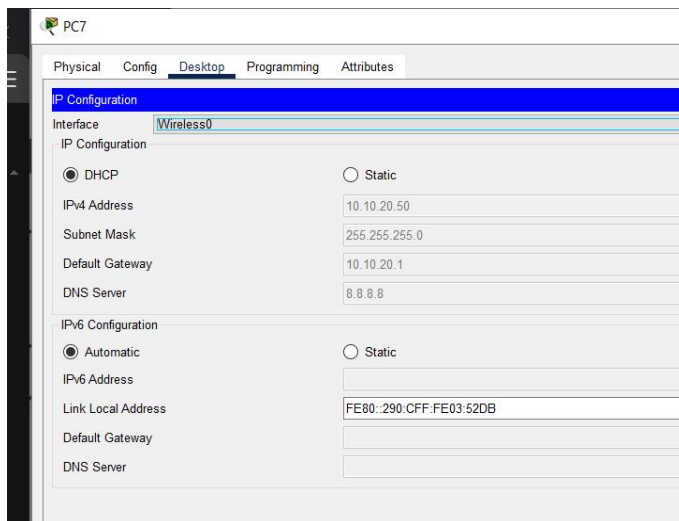
5. Security Rules

Rule	Mechanism	Effect
Guest isolation	Extended ACL GUEST_ISOLATION on R1 (Gi0/0.20 inbound)	Denies GUEST -> SERVER and GUEST -> MANAGEMENT; permits internet traffic.
NAT scope	Standard ACL NAT_ACL + 'ip nat inside source list ... overload'	Only EMPLOYEE and GUEST subnets are translated to the internet; SERVER/MANAGEMENT are not exposed.
Wireless security	WPA2-PSK/AES on AP1	Guest SSID requires authentication, preventing open/unauthenticated wireless access.

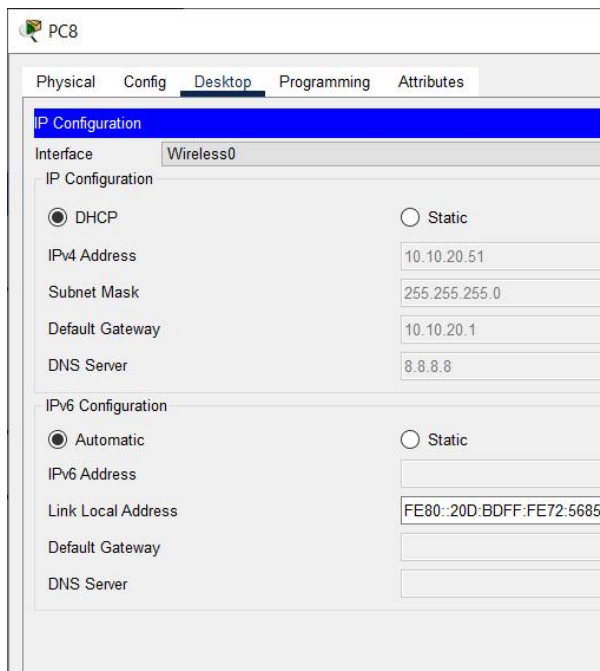
6. Verification Testing

Connectivity and isolation were verified directly via ping between hosts on different VLANs.

Test	Result
Employee (PC1/PC2) -> Server (10.10.30.10)	Success — replies received
Employee -> Internet (8.8.8.8)	Success — replies received
Guest (PC7/PC8, wireless) -> Server	Blocked — 'Destination host unreachable' (ACL working as intended)
Guest -> Management (10.10.99.x)	Blocked — 'Destination host unreachable'
Guest -> Internet	Success — replies received (guest is isolated from internal VLANs only, not the internet)



PC7 connected wirelessly to the guest SSID, received 10.10.20.50 via DHCP.



PC8 connected wirelessly to the guest SSID, received 10.10.20.51 via DHCP.

```

C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Reply from 8.8.8.8: bytes=32 time=48ms TTL=254
Reply from 8.8.8.8: bytes=32 time=44ms TTL=254
Reply from 8.8.8.8: bytes=32 time=26ms TTL=254
Reply from 8.8.8.8: bytes=32 time=23ms TTL=254

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 23ms, Maximum = 48ms, Average = 35ms

C:\>

```

Internet reachability confirmed end-to-end (ping 8.8.8.8, 0% loss) after NAT configuration.

```

R1>
R1>configure terminal
^
% Invalid input detected at '^' marker.

R1>enable
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ip access-list extended GUEST_ISOLATION
R1(config-ext-nacl)#deny ip 10.10.20.0 0.0.0.255 10.10.30.0 0.0.0.255
R1(config-ext-nacl)#deny ip 10.10.20.0 0.0.0.255 10.10.99.0 0.0.0.255
R1(config-ext-nacl)#permit ip any any
R1(config-ext-nacl)#exit
R1(config)#interface GigabitEthernet0/0.20
R1(config-subif)#ip access-group GUEST_ISOLATION in
R1(config-subif)#exit
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
R1#
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface GigabitEthernet0/1
R1(config-if)#ip address dhcp
R1(config-if)#

```

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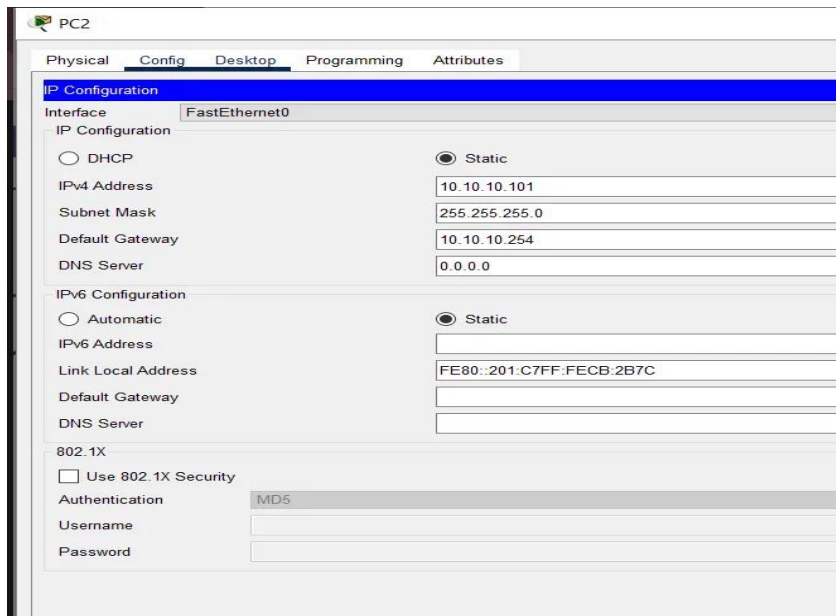
R1 CLI — GUEST_ISOLATION ACL and NAT sub-interface (ip nat inside) configuration on R1.

7. Fault Injection & Troubleshooting (5 Required Faults)

Each fault was deliberately injected, diagnosed from show-command / host evidence, fixed, and re-verified. Full before-and-after evidence follows.

Fault 1 — Wrong Default Gateway (Layer 3)

Injection: PC2 manually set with an incorrect gateway (10.10.10.254 instead of 10.10.10.1).



PC2 IP configuration showing the incorrect gateway 10.10.10.254 (injected fault).

Symptom: pings from PC2 to the server and to the internet both failed.

```
PC2
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.10.1

Pinging 10.10.10.1 with 32 bytes of data:

Reply from 10.10.10.1: bytes=32 time<1ms TTL=255
Reply from 10.10.10.1: bytes=32 time<1ms TTL=255
Reply from 10.10.10.1: bytes=32 time<1ms TTL=255
Reply from 10.10.10.1: bytes=32 time=2ms TTL=255

Ping statistics for 10.10.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>ping 10.10.30.10

Pinging 10.10.30.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.10.30.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>|
```

PC2 — ping 10.10.30.10 and ping 8.8.8.8 both time out with the wrong gateway set.

```
C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::201:C7FF:FECB:2B7C
    IPv6 Address . . . . .: ::
    IPv4 Address. . . . .: 10.10.10.101
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                   10.10.10.254

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address. . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                   0.0.0.0

C:\>
```

ipconfig confirms the faulty gateway (10.10.10.254) on PC2's FastEthernet0 interface.

Fix: gateway corrected back to 10.10.10.1. Re-test confirmed connectivity restored.

PC2

Physical Config Desktop Programming Attributes

Command Prompt

```
Subnet Mask.....: 0.0.0.0
Default Gateway.....: :
                  0.0.0.0

C:\>ping 10.10.30.10

Pinging 10.10.30.10 with 32 bytes of data:

Request timed out.
Reply from 10.10.30.10: bytes=32 time=1ms TTL=127
Reply from 10.10.30.10: bytes=32 time<1ms TTL=127
Reply from 10.10.30.10: bytes=32 time=1ms TTL=127

Ping statistics for 10.10.30.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 10.10.30.10

Pinging 10.10.30.10 with 32 bytes of data:

Reply from 10.10.30.10: bytes=32 time<1ms TTL=127
Reply from 10.10.30.10: bytes=32 time<1ms TTL=127
Reply from 10.10.30.10: bytes=32 time=11ms TTL=127
Reply from 10.10.30.10: bytes=32 time=1ms TTL=127

Ping statistics for 10.10.30.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 3ms

C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Request timed out.
Reply from 8.8.8.8: bytes=32 time<1ms TTL=254
Reply from 8.8.8.8: bytes=32 time<1ms TTL=254
Reply from 8.8.8.8: bytes=32 time=25ms TTL=254

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 25ms, Average = 8ms

C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Reply from 8.8.8.8: bytes=32 time<1ms TTL=254
Reply from 8.8.8.8: bytes=32 time=11ms TTL=254
Reply from 8.8.8.8: bytes=32 time=10ms TTL=254
Reply from 8.8.8.8: bytes=32 time=1ms TTL=254

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 5ms

C:\>
```

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PC2 — after correcting the gateway, ping 10.10.30.10 and ping 8.8.8.8 both succeed (0% loss).

Root cause: incorrect default gateway prevents any off-subnet routing — a Layer 3 addressing fault.

Fault 2 — Missing VLAN on Trunk (Layer 2)

Injection: VLAN 20 (GUEST) removed from the SW1<->R1 and SW2<->SW1 trunk allowed-list on both switches.


```

SW1>enable
SW1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#interface FastEthernet0/1
SW1(config-if)#switchport trunk allowed vlan remove 20
SW1(config-if)#end
SW1#
%SYS-5-CONFIG_I: Configured from console by console
show interfaces trunk
Port      Mode      Encapsulation  Status        Native vlan
Fa0/1     on        802.1q         trunking      1
Fa0/2     on        802.1q         trunking      1

Port      Vlans allowed on trunk
Fa0/1     10,30,99
Fa0/2     10,20,30,99

Port      Vlans allowed and active in management domain
Fa0/1     10,30,99
Fa0/2     10,20,30,99

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     none
Fa0/2     10,20,30,99
SW1#

```

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SW1 — 'switchport trunk allowed vlan remove 20' applied; VLAN 20 no longer in the trunk allowed list.

```
SW2
Physical Config CLI Attributes
IOS Command Line Interface

write memory
Building configuration...
[OK]
SW2#show ip interface brief | include Vlan99
Vlan99      10.10.99.12  YES manual up
SW2#

SW2 con0 is now available

Press RETURN to get started.

SW2>enable
SW2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
SW2(config)#interface FastEthernet0/1
SW2(config-if)#switchport trunk allowed vlan remove 20
SW2(config-if)#exit
SW2(config)#end
SW2#
%SYS-5-CONFIG_I: Configured from console by console
show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/1     on        802.1q         trunking    1

Port      Vlans allowed on trunk
Fa0/1     10,99

Port      Vlans allowed and active in management domain
Fa0/1     10,99

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     none
SW2#
```

SW2 — same fault applied; trunk allowed list now only shows 10,99.

Symptom: guest device PC7 could not renew its DHCP lease at all.

```

C:\>ipconfig /release
Port is not using DHCP.
C:\>ipconfig /renew

IP Address.....: 10.10.20.52
Subnet Mask.....: 255.255.255.0
Default Gateway...: 10.10.20.1
DNS Server.....: 8.8.8.8

C:\>ipconfig /release

IP Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway...: 0.0.0.0
DNS Server.....: 0.0.0.0

C:\>ipconfig /renew
DHCP request failed.

C:\>

```

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PC7 — DHCP release/renew fails ('DHCP request failed') because guest broadcast traffic has no path to R1.

Fix: VLAN 20 re-added to both trunks.

SW1

Physical Config CLI Attributes

IOS Command Line Interface

```

Fa0/2      10,20,30,99

SW1#

SW1 con0 is now available

Press RETURN to get started.


SW1>enable
SW1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#interface FastEthernet0/1
SW1(config-if)#switchport trunk allowed vlan add 20
SW1(config-if)#end
SW1#
%SYS-5-CONFIG_I: Configured from console by console
show interfaces trunk
Port      Mode      Encapsulation  Status        Native vlan
Fa0/1     on        802.1q         trunking      1
Fa0/2     on        802.1q         trunking      1

Port      Vlans allowed on trunk
Fa0/1     10,20,30,99
Fa0/2     10,20,30,99

Port      Vlans allowed and active in management domain
Fa0/1     10,20,30,99
Fa0/2     10,20,30,99

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     10,20,30,99
Fa0/2     10,20,30,99

SW1#

```

SW1 — VLAN 20 restored; show interfaces trunk now lists 10,20,30,99 on both Fa0/1 and Fa0/2.

SW2

Physical Config CLI Attributes

IOS Command Line Interface

```
Fa0/1      10,20,99
Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1      10,20,99

SW2#
SW2#

SW2 con0 is now available

Press RETURN to get started.

SW2>enable
SW2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW2(config)#interface FastEthernet0/1
SW2(config-if)#switchport trunk allowed vlan add 20
SW2(config-if)#end
SW2#
%SYS-5-CONFIG I: Configured from console by console
show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/1      on        802.1q         trunking    1

Port      Vlans allowed on trunk
Fa0/1      10,20,99

Port      Vlans allowed and active in management domain
Fa0/1      10,20,99

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1      10,20,99

SW2#
```

SW2 — VLAN 20 restored on the Fa0/1 trunk to SW1.

 PC7

Physical Config **Desktop** Programming Attributes

Command Prompt

```
C:\>ipconfig /release

IP Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway...: 0.0.0.0
DNS Server.....: 0.0.0.0

C:\>ipconfig /renew

IP Address.....: 10.10.20.52
Subnet Mask.....: 255.255.255.0
Default Gateway...: 10.10.20.1
DNS Server.....: 10.10.30.10
```

PC7 — DHCP renew now succeeds: 10.10.20.52 assigned, correct gateway and DNS.

Root cause: VLAN 20 missing from the trunk allowed-list blocked all guest traffic — a Layer 2 trunking fault.

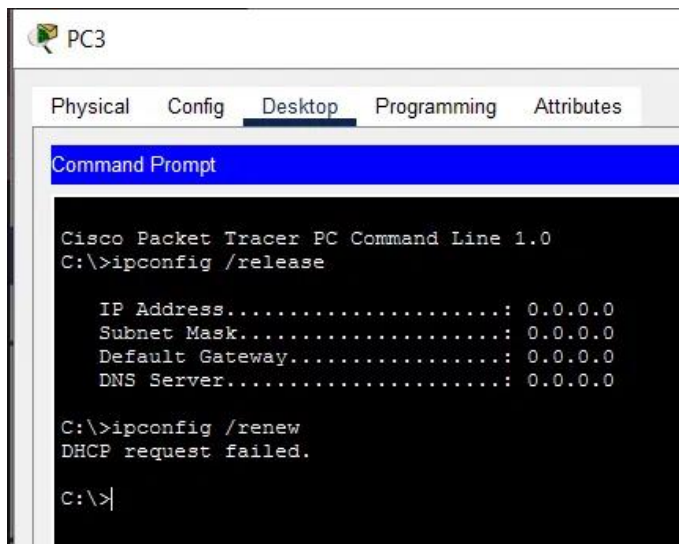
Fault 3 — DHCP Pool Exhausted (Layer 3 / DHCP)

Injection: R1's excluded-address range for the EMPLOYEE pool changed to 10.10.10.1-10.10.10.254, leaving almost no addresses to lease.

```
R1>enable
R1#show run | section dhcp
ip dhcp excluded-address 10.10.10.1 10.10.10.99
ip dhcp excluded-address 10.10.20.1 10.10.20.49
ip dhcp pool EMPLOYEE
 network 10.10.10.0 255.255.255.0
 default-router 10.10.10.1
 dns-server 8.8.8.8
ip dhcp pool GUEST
 network 10.10.20.0 255.255.255.0
 default-router 10.10.20.1
 dns-server 8.8.8.8
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#no ip dhcp excluded-address 10.10.10.1 10.10.10.99
R1(config)#ip dhcp excluded-address 10.10.10.1 10.10.10.254
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
```

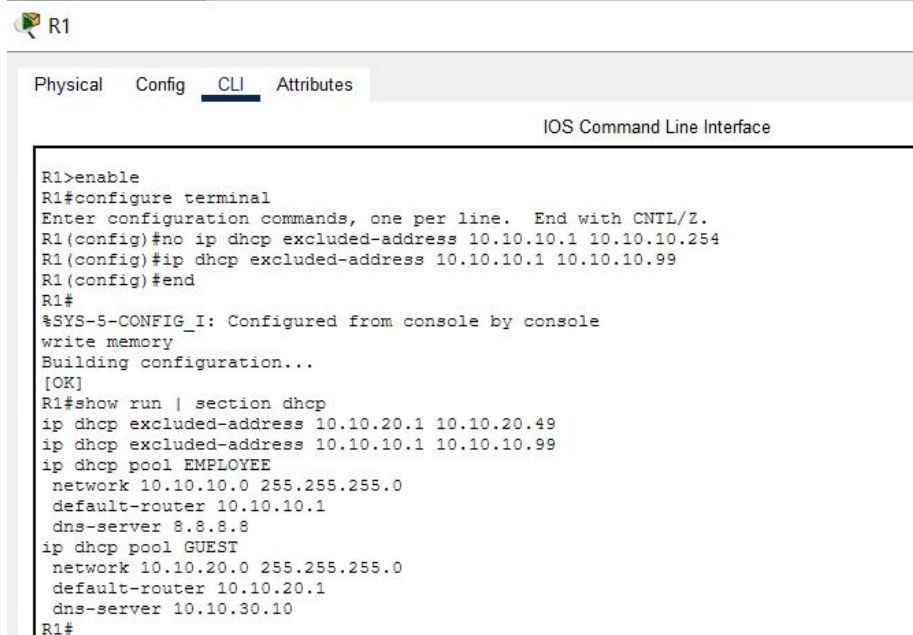
R1 — excluded-address range widened to 10.10.10.1-10.10.10.254 (injected fault).

Symptom: PC3, an existing employee host, failed to renew its DHCP lease.

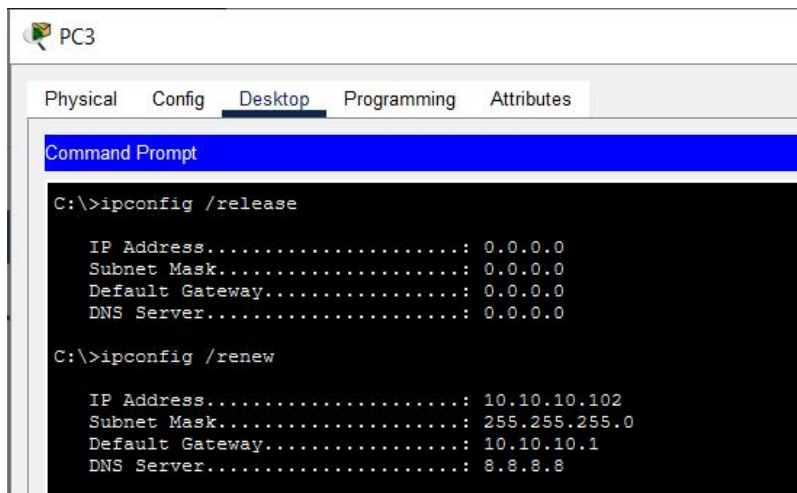


PC3 — DHCP renew fails ('DHCP request failed') with the pool exhausted.

Fix: excluded-address range corrected back to 10.10.10.1-10.10.10.99.



R1 — corrected excluded-address range confirmed via show run | section dhcp.



PC3

Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>ipconfig /release

IP Address. . . . .: 0.0.0.0
Subnet Mask. . . . .: 0.0.0.0
Default Gateway. . . . .: 0.0.0.0
DNS Server. . . . .: 0.0.0.0

C:\>ipconfig /renew

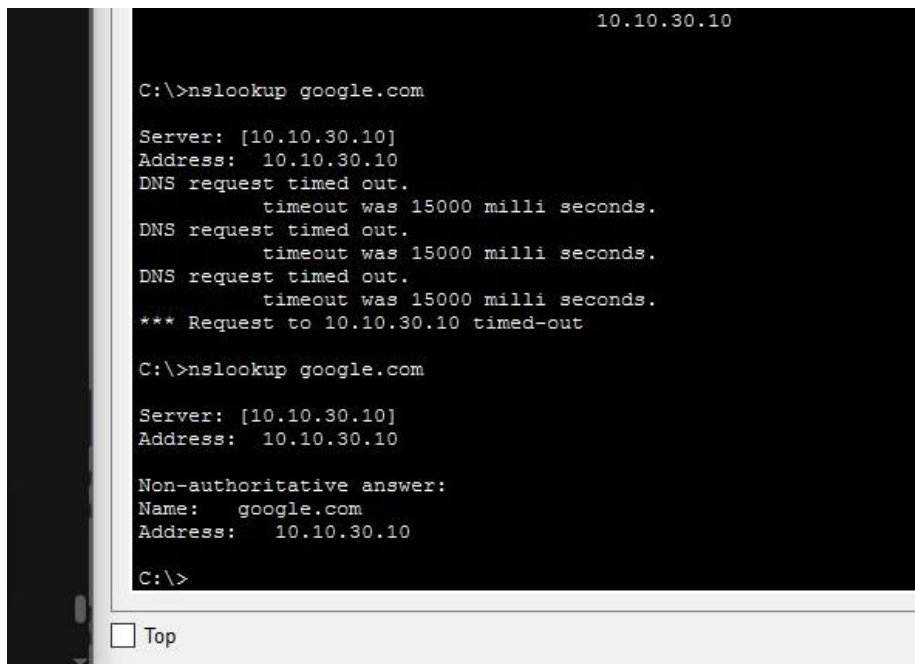
IP Address. . . . .: 10.10.10.102
Subnet Mask. . . . .: 255.255.255.0
Default Gateway. . . . .: 10.10.10.1
DNS Server. . . . .: 8.8.8.8
```

PC3 — DHCP renew now succeeds: 10.10.10.102 assigned with correct gateway.

Root cause: an overly broad excluded-address range starves the DHCP pool — a configuration fault, not a link/hardware fault.

Fault 4 — ACL Blocking DNS (Layer 4/7)

Injection: the GUEST_ISOLATION ACL contained a broad 'deny udp any any eq 53' rule, blocking all DNS resolution for the guest VLAN even though IP-based reachability still worked.



10.10.30.10

```
C:\>nslookup google.com

Server: [10.10.30.10]
Address: 10.10.30.10
DNS request timed out.
        timeout was 15000 milli seconds.
DNS request timed out.
        timeout was 15000 milli seconds.
DNS request timed out.
        timeout was 15000 milli seconds.
*** Request to 10.10.30.10 timed-out

C:\>nslookup google.com

Server: [10.10.30.10]
Address: 10.10.30.10

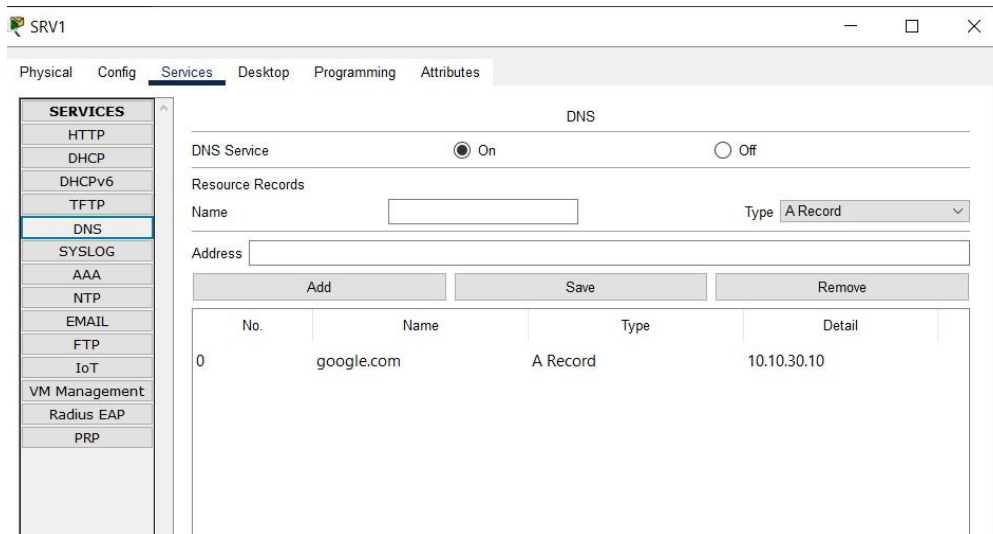
Non-authoritative answer:
Name:   google.com
Address: 10.10.30.10

C:\>
```

☐ Top

nslookup google.com times out repeatedly before finally resolving — intermittent/blocked DNS behaviour.

Fix: the broad deny was removed; an explicit permit for UDP/53 to an internal DNS host was added ahead of the guest-isolation deny rules, and SRV1 was configured as an internal DNS server (google.com -> 10.10.30.10) for verification.



SRV1 — DNS service enabled with an A record for google.com pointing to 10.10.30.10.

```
R1#enable
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ip access-list extended GUEST_ISOLATION
R1(config-ext-nacl)# no deny udp any any eq 53
R1(config-ext-nacl)# exit
R1(config)#end
R1#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
R1#enable
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(dhcp-config)#ip dhcp pool GUEST
R1(dhcp-config)# dns-server 10.10.30.10
R1(dhcp-config)# exit
R1(config)#end
R1#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
R1#show run | section dhcp pool GUEST
ip dhcp pool GUEST
 network 10.10.20.0 255.255.255.0
 default-router 10.10.20.1
 dns-server 10.10.30.10
R1#enable
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ip access-list extended GUEST_ISOLATION
R1(config-ext-nacl)# no deny ip 10.10.20.0 0.0.0.255 10.10.30.0 0.0.0.255
R1(config-ext-nacl)# 5 permit udp 10.10.20.0 0.0.0.255 host 10.10.30.10 eq 53
R1(config-ext-nacl)# 6 deny ip 10.10.20.0 0.0.0.255 10.10.30.0 0.0.0.255
R1(config-ext-nacl)# exit
R1(config)#end
R1#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
R1#show access-lists GUEST_ISOLATION
Extended IP access list GUEST_ISOLATION
  permit udp 10.10.20.0 0.0.0.255 host 10.10.30.10 eq domain
  deny ip 10.10.20.0 0.0.0.255 10.10.30.0 0.0.0.255
  deny ip 10.10.20.0 0.0.0.255 10.10.99.0 0.0.0.255 (8 match(es))
  permit ip any any (22 match(es))
R1#
```


R1 — GUEST_ISOLATION ACL updated: broad DNS deny removed, explicit permit for UDP/53 added; show access-lists confirms the corrected rule order and match counts.

Root cause: an ACL rule blocking UDP/53 breaks name resolution independent of underlying IP connectivity — an Application/Transport-layer fault.

Fault 5 — NAT ACL Missing Guest Subnet (Layer 3 / NAT)

Injection: the GUEST subnet (10.10.20.0/24) removed from NAT_ACL, so R1 no longer translated guest traffic for internet access.

```
R1#enable
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ip access-list standard NAT_ACL
R1(config-std-nacl)#no permit 10.10.20.0 0.0.0.255
R1(config-std-nacl)#exit
R1(config)#wnd
R1(config)#^
% Invalid input detected at '^' marker.

R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
R1#show access-lists NAT_ACL
Standard IP access list NAT_ACL
    permit 10.10.10.0 0.0.0.255 (32 match(es))

R1#show ip nat translations
R1#show ip nat statistics
Total translations: 0 (0 static, 0 dynamic, 0 extended)
Outside Interfaces: GigabitEthernet0/1
Inside Interfaces: GigabitEthernet0/0.10 , GigabitEthernet0/0.20
Hits: 45 Misses: 61
Expired translations: 32
Dynamic mappings:
R1#
```

R1 — NAT_ACL no longer permits 10.10.20.0/24; show ip nat translations shows zero active translations.

Symptom: guest devices lost internet access entirely while the employee VLAN was unaffected.

Fix: guest subnet re-added to NAT_ACL.

 R1

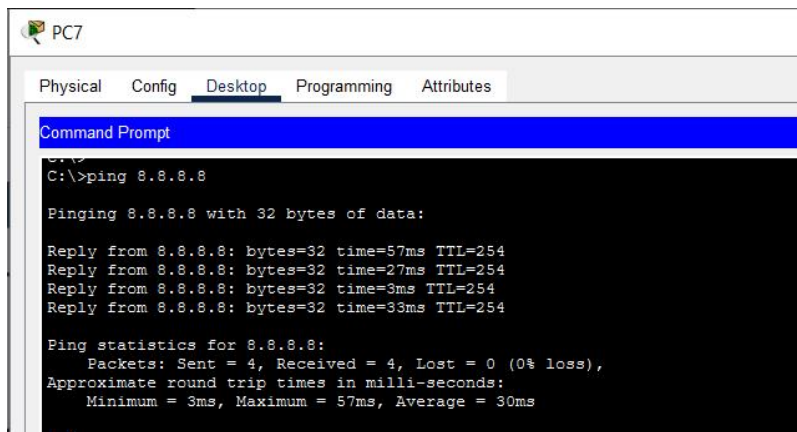
PhysicalConfigCLIAttributes

IOS Command Line Interface

```
R1>enable
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ip access-list standard NAT_ACL
R1(config-std-nacl)#permit 10.10.20.0 0.0.0.255
R1(config-std-nacl)#exit
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
R1#show access-lists NAT_ACL
Standard IP access list NAT_ACL
    permit 10.10.10.0 0.0.0.255 (32 match(es))
    permit 10.10.20.0 0.0.0.255 (8 match(es))

R1#
```

R1 — NAT_ACL corrected; show access-lists confirms both EMPLOYEE and GUEST subnets are now permitted for NAT.



The screenshot shows a window titled "PC7" with tabs for "Physical", "Config", "Desktop", "Programming", and "Attributes". The "Desktop" tab is active, displaying a "Command Prompt" window. The command prompt shows the execution of the command "C:\>ping 8.8.8.8". The output indicates that the ping was successful with 4 replies and 0% loss. The ping statistics for 8.8.8.8 are as follows:

```
C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Reply from 8.8.8.8: bytes=32 time=57ms TTL=254
Reply from 8.8.8.8: bytes=32 time=27ms TTL=254
Reply from 8.8.8.8: bytes=32 time=3ms TTL=254
Reply from 8.8.8.8: bytes=32 time=33ms TTL=254

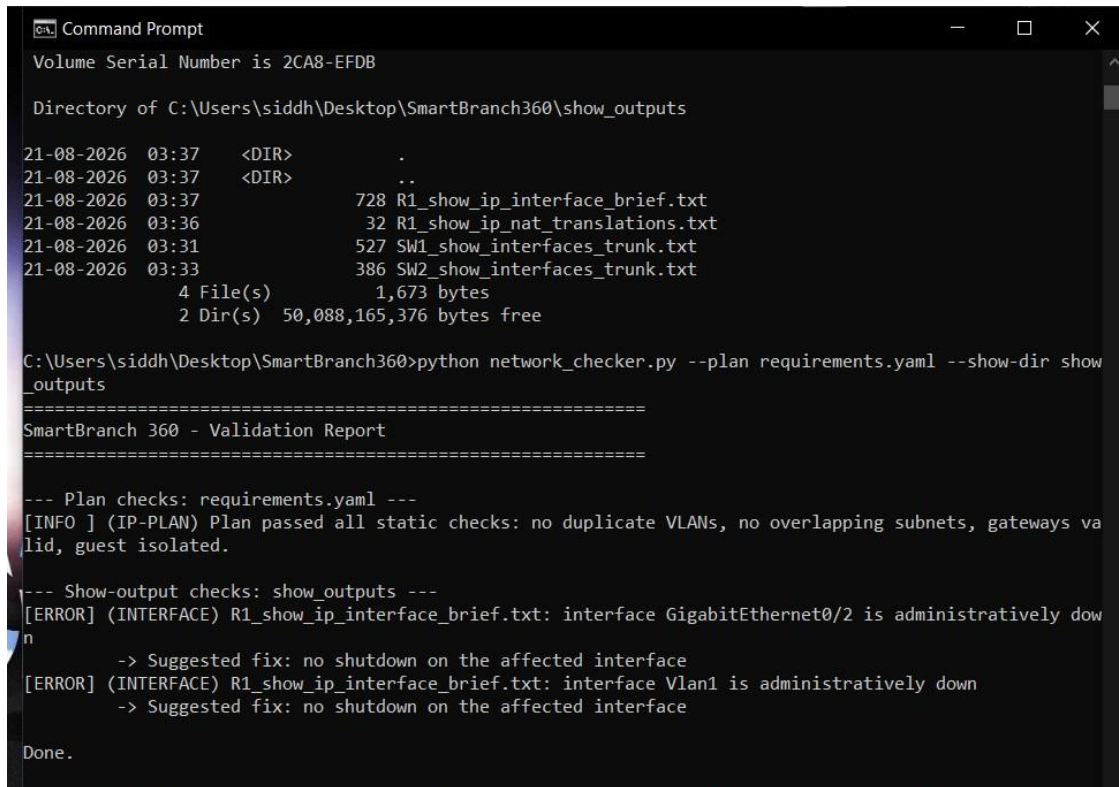
Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 57ms, Average = 30ms
```

PC7 (guest) — ping 8.8.8.8 now succeeds with 4/4 replies (0% loss), confirming internet access restored.

Root cause: NAT only translates traffic matched by NAT_ACL — omitting a subnet silently blocks its internet access even though routing is otherwise fine.

8. Python Assurance Tool

network_checker.py performs two kinds of validation: (1) static checks of the YAML VLAN/IP plan — duplicate VLAN IDs, overlapping subnets, gateway-in-subnet correctness, guest-isolation policy — and (2) dynamic checks of pasted show-command output — missing VLANs on trunks, interfaces down, duplicate-IP log entries, empty NAT translation tables. It was run against real show-command output captured from the built topology:



```

C:\Users\siddh\Desktop\SmartBranch360\show_outputs

21-08-2026 03:37 <DIR> .
21-08-2026 03:37 <DIR> ..
21-08-2026 03:37      728 R1_show_ip_interface_brief.txt
21-08-2026 03:36      32 R1_show_ip_nat_translations.txt
21-08-2026 03:31     527 SW1_show_interfaces_trunk.txt
21-08-2026 03:33     386 SW2_show_interfaces_trunk.txt
                4 File(s)      1,673 bytes
                2 Dir(s)  50,088,165,376 bytes free

C:\Users\siddh\Desktop\SmartBranch360>python network_checker.py --plan requirements.yaml --show-dir show_outputs
=====
SmartBranch 360 - Validation Report
=====

--- Plan checks: requirements.yaml ---
[INFO ] (IP-PLAN) Plan passed all static checks: no duplicate VLANs, no overlapping subnets, gateways valid, guest isolated.

--- Show-output checks: show_outputs ---
[ERROR] (INTERFACE) R1_show_ip_interface_brief.txt: interface GigabitEthernet0/2 is administratively down
-> Suggested fix: no shutdown on the affected interface
[ERROR] (INTERFACE) R1_show_ip_interface_brief.txt: interface Vlan1 is administratively down
-> Suggested fix: no shutdown on the affected interface

Done.
```

network_checker.py run in a real terminal against captured show-command output from R1, SW1, and SW2. Plan check passed cleanly; two unused interfaces (Gi0/2, Vlan1) were correctly flagged as administratively down — expected, since they are not part of this design (VLAN 99 is used for management instead of the default VLAN1).

9. Summary

Fault	OSI Layer	Status
1. Wrong default gateway	Layer 3	Diagnosed, fixed, re-verified
2. Missing VLAN on trunk	Layer 2	Diagnosed, fixed, re-verified
3. DHCP pool exhausted	Layer 3 / DHCP	Diagnosed, fixed, re-verified
4. ACL blocking DNS	Layer 4/7	Diagnosed, fixed, re-verified
5. NAT ACL missing subnet	Layer 3 / NAT	Diagnosed, fixed, re-verified

All required components are complete: working topology with VLAN segmentation, DHCP, NAT, guest isolation and wireless; 5 fault-injection/fix cycles with full evidence; and an independently working Python assurance tool validated against real captured output. Remaining submission steps: save the Packet Tracer file as SmartBranch360.pkt and record the 5-10 minute demo video walking through this report.